Markov Chains (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.03

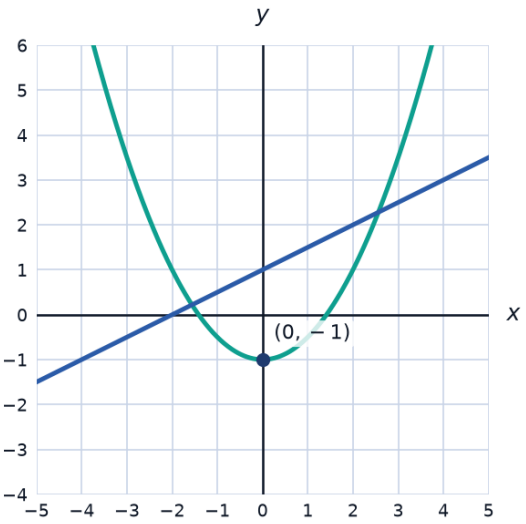
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Transition Matrices and State Vectors

Are $\mathbf{u} = (2; 0; 0)$ and $\mathbf{w} = (7; 0; 0)$ parallel? Show your complete algebraic work.

2 [Foundational] • Target: Multi-Step Transitions

Are $\mathbf{u} = (1; -1; 3)$ and $\mathbf{w} = (-2; 2; -5)$ parallel? Show your complete algebraic work.



3 [Foundational] • Target: Steady-State (Stationary) Distribution

Are $\mathbf{u} = (2; -4; 1)$ and $\mathbf{w} = (-4; 8; -1)$ parallel? Show your complete algebraic work.

4 [Foundational] • Target: Absorbing Markov Chains

Are $\mathbf{u} = (-2; -4; 0)$ and $\mathbf{w} = (6; 13; 0)$ parallel? Show your complete algebraic work.

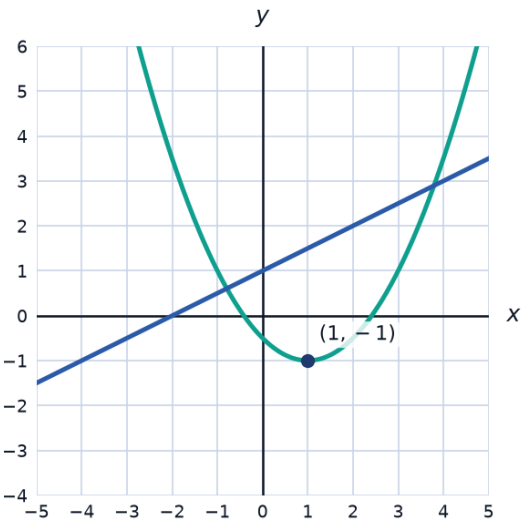
5 [Foundational] • Target: Transition Matrices and State Vectors

Are $\mathbf{u} = (2; 4; 0)$ and $\mathbf{w} = (-6; -12; 0)$ parallel? Show your complete algebraic work.

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6 [Foundational] • Target: Multi-Step Transitions

Are $\mathbf{u} = (0; -4; 0)$ and $\mathbf{w} = (0; 12; 0)$ parallel? Show your complete algebraic work.



7 [Foundational] • Target: Steady-State (Stationary) Distribution

Simplify $\frac{1-\cos^2\theta}{\sin\theta} \cdot \sec\theta$. .

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Absorbing Markov Chains

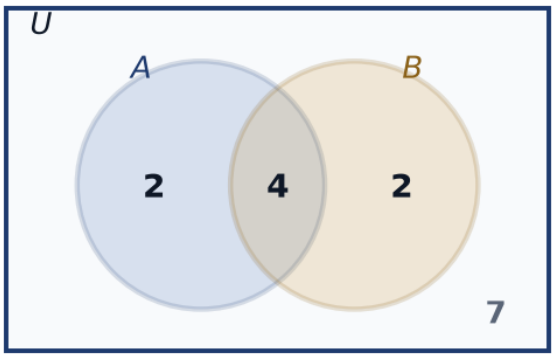
Line: $\mathbf{r} = (2; 2; -3) + t(-2; -2; 0)$. Plane: $-4x - 2z = 14$. Find the value of t at the point of intersection.

9 [Intermediate] • Target: Transition Matrices and State Vectors

Line: $\mathbf{r} = (4; -1; 3) + t(3; 2; -3)$. Plane: $-2x - 4y + 3z = 5$. Find the value of t at the point of intersection.

10 [Intermediate] • Target: Multi-Step Transitions

Line: $\mathbf{r} = (0; 2; 4) + t(2; 0; 1)$. Plane: $-x - 4y + 3z = 7$. Find the value of t at the point of intersection.



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11 [Intermediate] • Target: Steady-State (Stationary) Distribution

Find the distance from $P(-1,\ -13, 23)$ to the plane $-3x - 4y + 12z = -7$.

12 [Intermediate] • Target: Absorbing Markov Chains

Find the distance from $P(-28,\ -10, 11)$ to the plane $-6x - 2y + 3z = -24$.

13 [Intermediate] • Target: Transition Matrices and State Vectors

Find the distance from $P(-1, 2,\ -11)$ to the plane $-2x + 6y - 9z = -8$.

14 [Intermediate] • Target: Multi-Step Transitions

Find the distance from $P(-13,\ -33, 13)$ to the plane $-3x - 6y + 2z = 18$.

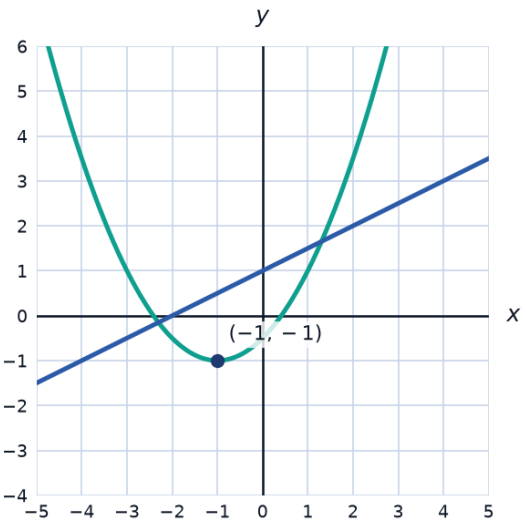
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find the distance from $P(25, 11, 16)$ to the plane $6x + 2y + 3z = -25$.

16 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 6.01 Matrices (AI HL)

Find the distance from $P(-15,\ -3,\ -8)$ to the plane $-6x - 2y - 3z = 22$.



17 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Line 1: $\mathbf{r} = (-2;\ -4; 2) + s(-2; 3;\ -1)$. Line 2: $\mathbf{r} = (0;\ -1; 4) + t(-4; 6;\ -2)$. Are the lines parallel? Show your complete algebraic work.

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18 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Line 1: $\mathbf{r} = (0;\ -1; 3) + s(-2; 4; 1)$. Line 2: $\mathbf{r} = (1; 3; 4) + t(-4; 8; 2)$. Are the lines parallel? Show your complete algebraic work.

19 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Line 1: $\mathbf{r} = (2; 2; 4) + s(0; 4;\ -2)$. Line 2: $\mathbf{r} = (-1;\ -5; 4) + t(0; 12;\ -6)$. Are the lines parallel? Show your complete algebraic work.

20 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Line 1: $\mathbf{r} = (-1; 5;\ -2) + s(2; 5;\ -3)$. Line 2: $\mathbf{r} = (-1; 4; 0) + t(4; 10;\ -6)$. Are the lines parallel? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Transition Matrices and State Vectors

In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1

b. 0

c. The number of states

d. $\frac{1}{n}$

22 [Foundational] • Target: Transition Matrices and State Vectors

Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. $[0.60, 0.40]$

b. $[0.55, 0.45]$

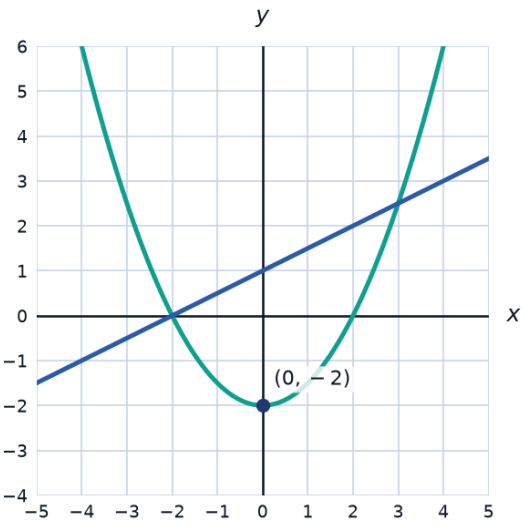
c. $[0.66, 0.34]$

d. $[0.48, 0.28]$

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23 [Foundational] • Target: Transition Matrices and State Vectors

What is the Markov property (memoryless property) in stochastic processes?



a. The future state depends only upon the current state, not on the sequence of states that preceded it

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

24 [Foundational] • Target: Multi-Step Transitions

How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

25 [Foundational] • Target: Multi-Step Transitions

If $T = [[0.5, 0.5], [0.5, 0.5]]$, evaluate T^2.

a. $[[0.5, 0.5], [0.5, 0.5]]$

b. $[[0.25, 0.25], [0.25, 0.25]]$

c. $[[1, 0], [0, 1]]$

d. $[[0.5, 0], [0, 0.5]]$

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26 [Foundational] • Target: Multi-Step Transitions

What does entry $(T^3)_{2,1}$ represent?

a. The probability of transitioning from State 2 to State 1 in exactly 3 steps

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

27 [Foundational] • Target: Steady-State (Stationary) Distribution

What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi} T = \boldsymbol{\pi}$ and $\sum \pi_i = 1$

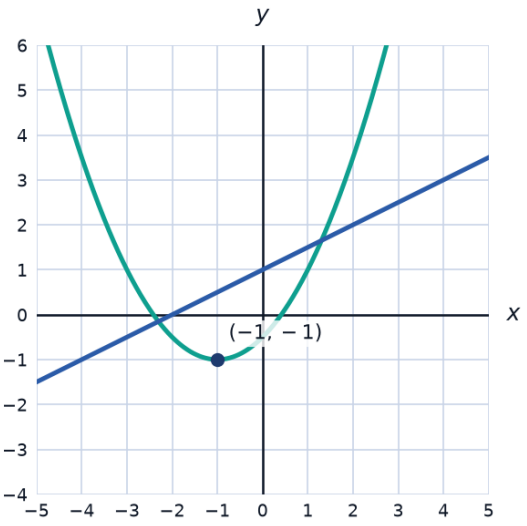
b. $T\boldsymbol{\pi} = \mathbf{0}$

c. $\boldsymbol{\pi} T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

28 [Foundational] • Target: Steady-State (Stationary) Distribution

Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.



a. $\boldsymbol{\pi} = [0.4, 0.6]$

b. $\boldsymbol{\pi} = [0.5, 0.5]$

c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Steady-State (Stationary) Distribution**

When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?

a. When the transition matrix is regular (some power T^k has strictly positive entries)

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

30 **[Intermediate]** • **Target: Absorbing Markov Chains**

What is an absorbing state in a Markov chain?

a. A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$)

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

31 **[Intermediate]** • **Target: Absorbing Markov Chains**

In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. The expected number of visits to each transient state before absorption

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

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32 [Intermediate] • Target: Absorbing Markov Chains

How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$

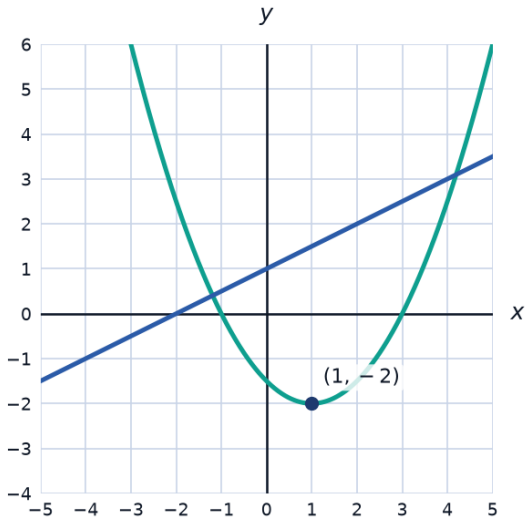
b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

33 [Intermediate] • Target: Transition Matrices and State Vectors

Line 1: $\mathbf{r} = (-3; 4; -4) + s(3; -5; -4)$. Line 2: $\mathbf{r} = (2; 4; 1) + t(6; -10; -8)$. Are the lines parallel?



a. They are the same line

b. Yes

c. They intersect at one point

d. No

34 [Intermediate] • Target: Multi-Step Transitions

Find the distance from $P(8, -5, 5)$ to the plane $6x - 2y + 3z = 24$.

a. 5

b. 4

c. 7

d. 2

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35 [Intermediate] • Target: Steady-State (Stationary) Distribution

Find the distance from the origin to the plane $3x + 4y - 12z = 52$.

a. 5

b. 6

c. 4

d. 2

36 [Intermediate] • Target: Absorbing Markov Chains

Find the distance from the origin to the plane $3x - 4y + 12z = 52$.

a. 1

b. 2

c. 4

d. 5

37 [Intermediate] • Target: Transition Matrices and State Vectors

Find the distance from the origin to the plane $-3x - 4y + 12z = 39$.

a. 3

b. 4

c. 1

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find the distance from the origin to the plane $-3x - 4y - 12z = 13$.

a. 3

b. 2

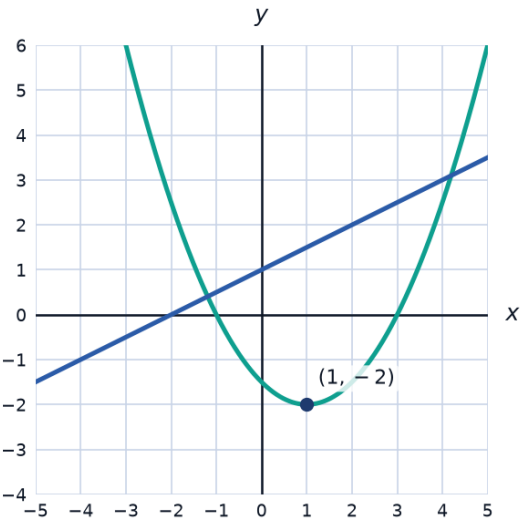
c. 4

d. 1

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39 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (-1; 5; -3)$ and $\mathbf{v} = (-5; -3; -6)$.



a. 4

b. 2

c. 11

d. 8

40 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (3; -4; 6)$ and $\mathbf{v} = (2; -2; -6)$.

a. -22

b. -23

c. -30

d. -33

41 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 1.04 Partial Fractions

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (-6; 4; -1)$ and $\mathbf{v} = (-6; 4; -1)$.

a. 42

b. 53

c. 56

d. 50

42 [Challenging] • Target: Multi-Step Transitions • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{u} = (6; 6; 5)$ and $\mathbf{v} = (-6; -2; -2)$.

a. -54

b. -69

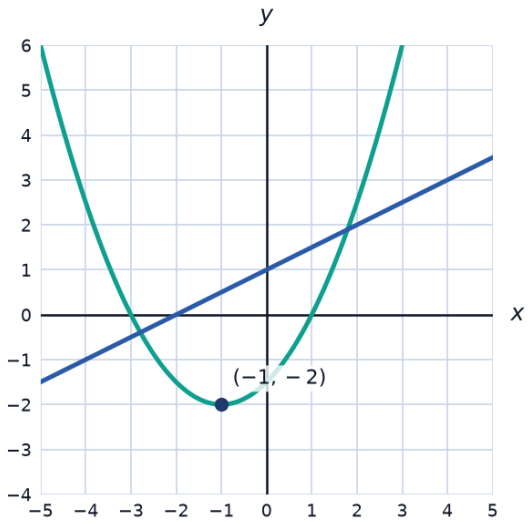
c. -51

d. -58

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43 [Challenging] • Target: Steady-State (Stationary) Distribution • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the distance from the origin to the plane $3x + 6y + 2z = 28$.



a. 7

b. 3

c. 2

d. 4

44 [Challenging] • Target: Absorbing Markov Chains • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find the distance from the origin to the plane $4x - 4y + 7z = 36$.

a. 4

b. 1

c. 7

d. 2

45 [Challenging] • Target: Transition Matrices and State Vectors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the distance from the origin to the plane $-4x - 4y + 7z = 18$.

a. 2

b. 5

c. 4

d. 1

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